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BATCH: DEVOPS B2

LAB EXERCISE 16- CREATING A CONFIGMAP FROM A FOLDER

IN THIS LAB EXERCISE, YOU WILL LEARN HOW TO CREATE A CONFIGMAP IN KUBERNETES FROM A FOLDER CONTAINING MULTIPLE FILES. THIS ALLOWS YOU TO MANAGE MULTIPLE CONFIGURATION FILES AS A SINGLE CONFIGMAP.

STEP 1: SET UP KUBERNETES CLUSTER

ENSURE YOU HAVE ACCESS TO A KUBERNETES CLUSTER. YOU CAN USE A LOCAL SETUP WITH MINIKUBE, KIND, OR USE A CLOUD-BASED KUBERNETES SERVICE.

STEP 2: PREPARE THE FOLDER WITH MULTIPLE FILES

CREATE A DIRECTORY NAMED CONFIGMAP-FOLDER:

```
MKDIR CONFIGMAP-FOLDER  
CD CONFIGMAP-FOLDER/
```

CREATE MULTIPLE CONFIGURATION FILES INSIDE THE DIRECTORY:

CONFIG1.TXT WITH CONTENT:

```
KEY1=VALUE1
```

CONFIG2.TXT WITH CONTENT:

```
KEY2=VALUE2
```

CONFIG3.TXT WITH CONTENT:

```
KEY3=VALUE3
```

VERIFY THE DIRECTORY STRUCTURE:

```
LS CONFIGMAP-FOLDER/  
CD ..
```

YOU SHOULD SEE THE FILES CONFIG1.TXT, CONFIG2.TXT, AND CONFIG3.TXT.

STEP 3: CREATE A CONFIGMAP FROM THE FOLDER

CREATE A CONFIGMAP NAMED FOLDER-CONFIGMAP FROM THE FOLDER:

```
KUBECTL CREATE CONFIGMAP FOLDER-CONFIGMAP --FROM-FILE=CONFIGMAP-FOLDER/  
configmap/folder-configmap created
```

VERIFY THE CONFIGMAP:

```
KUBECTL GET CONFIGMAP FOLDER-CONFIGMAP -O YAML
```

```

apiVersion: v1
data:
  config1.txt: |
    key1=value1
  config2.txt: |
    key2=value2
  config3.txt: |
    key3=value3
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T06:02:48Z"
  name: folder-configmap
  namespace: default
  resourceVersion: "6555"
  uid: 3819543c-cca7-4066-84c7-87cfdc53fafa

```

YOU SHOULD SEE THE FOLDER-CONFIGMAP WITH THE CONTENTS OF ALL THE FILES IN THE CONFIGMAP-FOLDER.

STEP 4: USE THE CONFIGMAP IN A POD

CREATE A FILE NAMED **POD-FOLDER-CONFIGMAP.YAML** WITH THE FOLLOWING CONTENT:

```

APIVERSION: V1
KIND: POD
METADATA:
  NAME: CONFIGMAP-POD
SPEC:
  CONTAINERS:
  - NAME: NGINX
    IMAGE: NGINX
    ENVFROM:
    - CONFIGMAPREF:
      NAME: FOLDER-CONFIGMAP

```

```

apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
      name: folder-configmap

```

IN THIS MANIFEST:

- THE POD MOUNTS THE FOLDER-CONFIGMAP AT /ETC/CONFIG.
- THE BUSYBOX CONTAINER CONCATENATES AND PRINTS THE CONTENTS OF THE FILES.

APPLY THE MANIFEST TO CREATE THE POD:

```
KUBECTL APPLY -F POD-FOLDER-CONFIGMAP.YAML
```

CHECK THE STATUS OF THE POD:

```
KUBECTL GET PODS
```

NAME	READY	STATUS	RESTARTS	AGE
configmap-pod	1/1	Running	0	14s

VIEW THE POD LOGS TO SEE THE CONTENT OF THE MOUNTED FILES:

```
KUBECTL EXEC -IT CONFIGMAP-POD -- PRINTENV
```

```
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
TERM=xterm
config3.txt=key3=value3

config1.txt=key1=value1

config2.txt=key2=value2

NODEPORT_SERVICE_PORT_80_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PROTO=tcp
NODEPORT_SERVICE_PORT_80_TCP_PORT=80
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_PORT=tcp://10.96.0.1:443
NODEPORT_SERVICE_SERVICE_PORT=80
NODEPORT_SERVICE_PORT_80_TCP_ADDR=10.101.115.216
KUBERNETES_SERVICE_PORT=443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
NODEPORT_SERVICE_SERVICE_HOST=10.101.115.216
NODEPORT_SERVICE_PORT=tcp://10.101.115.216:80
NODEPORT_SERVICE_PORT_80_TCP=tcp://10.101.115.216:80
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT_443_TCP_PORT=443
NGINX_VERSION=1.29.5
NJS_VERSION=0.9.5
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
HOME=/root
```

YOU SHOULD SEE THE CONTENTS OF CONFIG1.TXT, CONFIG2.TXT, AND CONFIG3.TXT PRINTED.

STEP 5: CLEAN UP

AFTER COMPLETING THE EXERCISE, CLEAN UP THE RESOURCES CREATED:

```
KUBECTL DELETE POD FOLDER-CONFIGMAP-POD
```

```
KUBECTL DELETE CONFIGMAP FOLDER-CONFIGMAP
```

RM -R CONFIGMAP-FOLDER